

JIM WOODRUFF LOCK AND DAM AND LAKE SEMINOLE PERTINENT DATA

GENERAL

Damsite, miles above mouth of Apalachicola River	106.3
Drainage from Chattahoochee River, square miles	8,708
Drainage from Flint River, square miles	8,456
Total drainage area above Jim Woodruff, square miles	17,164
Total drainage area in Apalachicola Basin, square miles	19,573

RESERVOIR

Length at elevation 77.0 feet NGVD29, miles	
Flint River / Chattahoochee River to George W. Andrews	47 / 46.8
Minimum operating pool elevation, feet NGVD29	76.5
Normal operating pool elevation, feet NGVD29	77.0
Maximum operating pool elevation, feet NGVD29	77.5
Emergency drawdown elevation, feet NGVD29	76.0
Absolute maximum pool elevation, feet NGVD29 (non-flood conditions)	77.8
Area at pool elevation 77.0 feet NGVD29 – acres	37,500
Total volume at elevation 77.0 feet NGVD29 – acre-feet	367,318
Shoreline miles at elevation 77.0 NGVD29 – miles	532

GATED SPILLWAY

Total length, including end pier – feet	766
Crest Elevation – feet NGVD29	48.0

FIXED-CREST SPILLWAY

Total length – feet	1,634
Crest Elevation – feet NGVD29	79.0

EARTH OVERFLOW DIKE

Total length (crest elevation 85.0 feet NGVD29) - feet	2,130
Total length of transition (elev. 85.0 to 107.0 feet NGVD29) - feet	690
Top width – feet	25

POWER PLANT

Generating capacity (declared*) MW (3 units @ 14.45)	43.35
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NAVIGATION LOCK

Distance center to center of gate pintles – feet	505
Maximum lift – feet	33

* Declared generating capacity is defined as the plant's operational capacity declared on a weekly basis to the power marketing agency. The value may vary slightly from week to week depending on factors such as head and cooling capabilities; values shown are the nominal values reported.

STREAM FLOW

Drainage area at dam site – square miles	17,164
Period of continuous record (Chattahoochee gage)	1923-2015
Average annual flow for period of record – cfs	21,489
Minimum monthly flow for period of record – cfs (Aug 1988)	4,750
Maximum monthly flow at damsite (Mar 1929) – cfs	171,632
Discharge at bankfull – cfs	77,000
Maximum design flood of record (March-April 1929)	
Natural peak inflow – cfs	303,400
Regulated peak outflow – cfs	285,500
Regulated peak headwater – feet NGVD29	82.34
Peak tailwater elevation – feet NGVD29	80.40
Spillway design flood	
Peak inflow to full reservoir – cfs	781,100
Regulated peak outflow – cfs	690,700
Regulated peak headwater – feet NGVD29	96.73
Peak tailwater elevation – feet NGVD29	96.10
Total rainfall (Spillway design flood) – inches	
Chattahoochee basin	14.2
Flint basin	13.4
Entire basin above dam	13.8
Standard project flood	
Peak inflow to full reservoir – cfs	781,100
Regulated peak outflow – cfs	690,700
Regulated peak headwater – feet NGVD29	86.39
Peak Tailwater elevation – feet NGVD29	84.95

GENERAL HYDRAULIC DATA

Conditions	Total discharge cfs	Tailwater elev. Feet NGVD29	Head water elev. Feet NGVD29	Swell- head ft	Average interval between occurrences, years
Pool begins to rise above normal due to flood inflows	203,600	75.3	77.0	1.70	25
Flow begins over fixed-crest spill-way	228,000	77.1	79.0	1.90	47
Flow begins over lock	278,000	80.0	82.0	2.00	Over 100
Maximum flood of record	285,500	80.40	82.34	1.96	Over 100
Flow begins over dike	345,000	83.3	85.0	1.70	Over 100
Flow begins over saddle, right bank	519,000	90.1	91.0	0.90	Over 1,000
Spillway-design flood	690,700	96.73	96.10	0.63	Over 1,000

RESERVOIR

Minimum operating pool elevation, feet NGVD29	76.5
Normal operating pool elevation, feet NGVD29	77.0
Maximum operating pool elevation, feet NGVD29	77.5
Emergency drawdown elevation, feet NGVD29	76.0
Absolute maximum pool elevation, feet NGVD29 (non-flood conditions)	77.8
Area at pool elevation 77.0 feet NGVD29– acres	37,500
Area acquired in fee simple – acres	60,072
Area acquired by easement with right to inundate – acres	6,516
Area in river bed – acres	4,000
Total area within taking line – acres	70,588
Total reservoir volume at elevation 77.0 feet NGVD29 – acre-feet	367,318
Shoreline miles at elevation 77.0 feet NGVD29 – miles	532
Reservoir length at elevation 77.0 feet NGVD29– river miles	
Chattahoochee River to George W. Andrews L&D	46.8
Flint River	47.0

LOCK

Nominal size of chamber – feet	82 x 450
Distance center to center of gate pintles – feet	505
Maximum lift – feet	33
Elevation of upper miter sill – feet NGVD29	54.0
Elevation of lower miter sill – feet NGVD29	30.0
Elevation of top of lock walls – feet NGVD29	82.0
Elevation of top of guide walls – feet NGVD29	82.0
Elevation of top of guard walls – feet NGVD29	82.0
Freeboard on upper guide and guard walls when lock becomes inoperative – feet	82.0
Freeboard on lower guide and guard walls when lock becomes inoperative – feet	2.8
Percent of time lock is inoperative	75
Height of upper gate	28' 4"
Height of lower gate	52' 4"
Size of culvert at valves – feet	10 x 10
Depth of submergence of valves by lower pool – feet	7.0
Type of emergency dams	Stop-log
Type of operating machinery	Hydraulic
Swellhead when lock is over topped – feet	2.0

LOCK OPERATING CHARACTERISTICS – FILLING

1.5-minute valve-opening time	
Total time required to fill lock chamber, starting with valves closed – minutes	6.31
Maximum rate of rise – fpm	10.25
Maximum flow into chamber – cfs	7,010
Filling coefficient	0.88
6-minute valve-opening time	
Total time required to fill lock chamber, starting with valves closed – minutes	8.58
Maximum rate of rise – fpm	6.23
Maximum flow into chamber – cfs	4,260
Filling coefficient	0.88

LOCK OPERATING CHARACTERISTICS – EMPTYING

1.5-minute valve-opening time	
Total time required to empty lock chamber, starting with valves closed – minutes	7.25
Maximum rate of fall – fpm	8.94
Maximum flow from chamber – cfs	6,110
Emptying coefficient	0.75
6-minute valve-opening time	
Total time required to empty lock chamber, starting with valves closed – minutes	9.52
Maximum rate of fall – fpm	5.75
Maximum flow from chamber – cfs	3,390
Emptying coefficient	0.75

FIXED-CREST SPILLWAY

Total length – feet	1,634
Net length – feet	1,584
Elevation of crest – feet NGVD29	79.0
Elevation of bucket – feet NGVD29	69.39

GATED SPILLWAY

Total length, including end pier – feet	766
Net length – feet	640
Elevation of crest – feet NGVD29	48.0
Number of piers including end pier	16
Width of piers – feet	
End pier	6
Other piers	8
Height of piers above spillway crest – feet	59
Type of gates	Vertical lift (split leaf)
Number of gates	16
Length of gates – feet	40
Height of gates – feet	30.5
Weight of top leaf – lbs	47,444
Weight of bottom leaf – lbs	66,402
Method of operating gates	Gantry Crane
Number of gantry cranes provided	2*
*one of these cranes will operate powerhouse intake gates also	
Capacity of gantry crane – tons	75
Maximum discharge capacity (pool elev. 77.0) – cfs	203,600
Elevation of top of gates in closed position – feet NGVD29	78.5
Elevation of access bridge – feet NGVD29	107.0
Elevation of apron – feet NGVD20	20.0 (max)
Thickness of apron – feet	3 (min)

EARTH OVERFLOW DIKE

Total length (crest elevation 85.0 feet NGVD29) – feet	2,130
Total length of transition (elev. 85.0 to elev. 107.0 feet NGVD29) – feet	690
Top width – feet	25
Side Slopes	18" grouted riprap on 6" gravel base
Maximum swellhead when dike is overtopped – feet	1.8
Recurrence interval of flood which will overtop dike – years	Over 100
Freeboard, top of dike above normal upper pool – feet	8

POWER PLANT

Overload capacity at normal operating pool - kW	43,350
Turbine discharge at normal pool (elev. 77.0) – cfs	
1 unit operating	5,500
2 units operating	12,000
3 units operating	18,300
Tailwater elevations – feet NGVD29	
1 unit operating	41.5
2 units operating	45.3
3 units operating	48.4
Net head, full load – feet	30.5
Minimum head for generation – feet	9
Percent of time minimum head equaled or exceeded	99.2
Powerhouse	
Length – feet	258.7
Width – feet	122.0
Type of construction	Reinforced concrete
Type of turbines	Kaplan
Number of units	3
Spacing of units – feet	65
Installed capacity (3@ 14,450) – kW	43,345
Capacity of turbines at 75 rpm and rated head (32.5') – hp	20,000
Elevation centerline of distributor – feet NGVD29	52.0
Elevation centerline of intake – feet NGVD29	53.5
Elevation low point of draft tube – feet NGVD29	4.0
Generator rating, kW (.09 p.f., 16.055 kVA, 3-phase, 60 cycles, 75 rpm)	14,450
Station auxiliary power	
Source	2 taps from 13.8 kv bus
Number of 3-phase transformers	2
Emergency gas generator – kW	400
Type of intake gates	Double leaf, vertical lift
Number of intake gates, each unit	2 (4 leaves)
Number of intake gates, provided	4 (8 leaves)
Size of intake gates	23'9" x 30'7"
Method of operating intake gates	75 ton gantry crane*
(*This crane is one of the two provided to operate spillway gates.)	
Type of draft tube gate	Vertical leaf (slide)
Number of draft tube gates, each unit	2 (4 leaves)
Number of draft tube gates, provided	4 (8 leaves)
Method of operating draft tube gates	20-ton gantry crane

POWER PLANT (Cont'd)

Switchyard and parking areas

Location	East of powerhouse
Elevation – feet NGVD29	107.0
Width of section – feet	260
Length of section – feet	491
Slope protection	18" riprap on 6" gravel base
Side slopes ground surface to elevation 85.0 feet NGVD29	1 on 4
Side slopes, elev. 85.0 to elev. 107.0 feet NGVD29	1 on 3
Drawdown for pondage (normally elev 77.5 to 76.5 feet NGVD29) – ft	1
Volume for pondage – acre feet	37,500
Plant output – kW	
At full load	43,350
At minimum generating head	6,000
Est. average annual energy during design – kWh	238,090,000
Actual average annual production (1957-2015) – kWh	209,445,000